Lattice Distortions and Ion Conduction Mechanisms in NASICON Solid Electrolytes

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Abstract

1 Introduction

The global transition to renewable energy demands safe, cost-effective, and high-energy-density electrochemical storage. Solid-state sodium batteries, in which a dense ceramic ion conductor replaces the flammable liquid electrolyte, offer intrinsic safety advantages and access to earth-abundant sodium resources[1,2]. Among the candidate sodium solid-state electrolytes (SSEs), materials of the Na superionic conductor (NASICON) family, with general formula $Na_xM_2(AO_4)_3$, combine high ionic conductivity with wide electrochemical stability windows[3,4]. Yet even the best NASICONs achieve room-temperature conductivities of order 10^{-3} S cm^{-1} [5,4]—roughly an order of magnitude below typical liquid electrolytes[6]. Closing this gap requires understanding how chemical composition governs the atomic-scale dynamics of ion transport.

The NASICON framework consists of corner-sharing MO_6 octahedra and AO_4 tetrahedra arranged in rhombohedral ($R\bar{3}c$) symmetry, creating a three-dimensional network of Na-ion conduction channels[4]. Na ions occupy two crystallographically distinct sites—Na1 (6b) and Na2 (18e)—connected through triangular geometric bottlenecks whose dimensions are set by the framework chemistry[6,4]. Systematic computational and experimental campaigns have established that ionic conductivity is optimised by tuning the Na content close to three per formula unit, the average metal-site cation radius to ~0.72 Å, and the silicate-to-phosphate polyanion ratio[4,7]. A high-entropy strategy—mixing multiple metal species on the M site—was subsequently shown to boost conductivity by orders of magnitude through local structural distortions that flatten the Na-site energy landscape and promote correlated ion migration[8]. These distortions are predicted to control the crossover between single-ion hopping (SIH), favoured when the 6b–18e site-energy difference is large, and occupancy-conserved hopping (OCH), which dominates when site energies overlap[6].

Distinguishing between these mechanisms is central to rationalising conductivity trends and to formulating predictive design rules for next-generation NASICON electrolytes.

Despite rapid progress in computational screening and bulk transport measurements, directly observing the lattice-scale dynamics that underpin SIH and OCH remains an open challenge. Electrochemical impedance spectroscopy and *ab initio* molecular dynamics probe macroscopic or simulated transport[4,9], while quasielastic neutron scattering (QENS) and nuclear magnetic resonance (NMR) relaxometry access diffusion on their respective characteristic time (1 ps–1 ns/1 µs–1 s) and length (1-10 Å/bulk) scales [10,11]. None of these techniques, however, provides real-time, momentum-resolved sensitivity to collective lattice fluctuations during ion conduction. X-ray photon correlation spectroscopy (XPCS) fills this gap: by tracking the temporal decorrelation of coherent Bragg speckle patterns, XPCS measures equilibrium and non-equilibrium dynamics at the atomic spatial scale on timescales from 10^{-2} to 10^{3} s[12,13]. The advent of fourth-generation synchrotron sources now delivers the coherent flux necessary to perform Bragg XPCS on micrometre-scale crystallites embedded in polycrystalline or composite samples[13].

Here we report Bragg-peak XPCS measurements on a systematic series of NASICON compositions in which the metal species M is varied to tune the Na-site energy landscape while all other compositional parameters are held as constant as practicable. Performed at room temperature on the (116), (104) and (113) Bragg reflections, the intensity autocorrelation functions $g^{(2)}(\mathbf{q}, \tau)$ reveal composition-dependent relaxation timescales and stretched-exponential exponents β that correlate with the predicted SIH–OCH crossover. Compositions with a larger $6b$–$18e$ energy difference exhibit faster decorrelation and higher β, consistent with the spatially homogeneous lattice relaxation expected for single-ion hopping, whereas compositions in the OCH regime display slower, more heterogeneous dynamics. Two-time correlation analysis further uncovers non-stationary fluctuations in select compositions, pointing to cooperative rearrangement events during ion transport. These results provide nanoscale dynamic evidence that directly links local lattice distortions to the ion-conduction mechanism in NASICONs, and establish Bragg XPCS as a characterisation tool for solid-state electrolytes and ionic conductors more broadly.

2 Results

3 Discussion

4 Methods

5 Acknowledgments

References

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